Annals of Operations Research (2025) 354:103–142
https://doi.org/10.1007/s10479-024-06348-z

ORIGINAL RESEARCH

Explainable artificial intelligence to improve the resilience of perishable product supply chains by leveraging customer characteristics

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Received: 25 July 2023 / Accepted: 7 October 2024 / Published online: 26 October 2024
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Abstract
Optimizing costs and profits while meeting customer demand is a critical challenge in the development of perishable supply chains. Customer-centric demand forecasting addresses this challenge by considering customer characteristics when determining inventory levels. This study proposes a solution framework comprising two steps: (a) segmentation using customer characteristics and (b) demand forecasting for each segment using transparent and responsible artificial intelligence techniques. We employed k-means, hierarchical clustering, and explainable AI (XAI) to segment, model, and compare several machine-learning techniques for demand forecasting. The results showed that support vector regression outperformed the autoregressive models. The results also showed that the two-step segmentation and demand forecasting process using hierarchical clustering and LSTM outperforms (Weighted average RMSE across segments = 61.57) the conventional single-step unsegmented forecasting process (RMSE overall data = 238.18). The main implication of this study is the demonstration of XAI in enhancing transparency in machine learning and an improved method for reducing forecasting errors in practice, which can strengthen the supply chain resilience for perishable products.

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Keywords Explainable artificial intelligence · Supply chain resilience · Inventory management · Demand forecasting · Perishable food products

Abbreviations

AR	Autoregression
AI	Artificial Intelligence
ANN	Artificial Neural Networks
COVID-19	Coronavirus Disease 2019
CAGR	Compound Annual Growth Rate
DL	Deep Learning
EDA	Exploratory Data Analysis
FMCG	Fast-Moving Consumer Goods
IoT	Internet of Things
IEEE	Institute of Electrical and Electronics Engineers
xDNN	Explainable Deep Neural Networks
LSTM	Long Short-Term Memory
LIME	Locally Interpretable Model Agnostic Explanation
ML	Machine Learning
MSE	Mean Squared Error
SHAP	Shapley Additive Explanation
SVM	Support Vector Method
RAM	Random-Access Memory
RMSE	Root Mean Square Error
VAR	Vector Autoregressive
UK	United Kingdom
UN	United Nations
XAI	Explainable AI

1 Introduction

A supply chain is a network of interconnected operations that produce consumer-beneficial products or services (Faruquee et al., 2021). Purchasing raw materials, work-in-progress (involving intermediate forms of finished items), and distributing finished goods to end users comprise the operations involved (Ali & Govindan, 2021). Across the supply chain, there should be a proper flow of items and information from consumers to suppliers, and vice versa, to meet the requirement of delivering the correct set of products in the correct quantity and at the correct time to the correct customer (Oluwaseyi et al., 2017).

Inventory management is one of the most important supply chain tasks and presents an opportunity to reduce expenses. Therefore, focusing on this work is essential to enhance supply chain performance (Creazza et al., 2021; Faruquee et al., 2021). With the emergence of Artificial Intelligence (AI) in recent years, there have been enormous opportunities to deploy such technologies in inventory management (Sharma et al., 2020). Inventory management is an ongoing inventory planning, organization, and control process to minimize inventory investment and balance supply and demand (Singh & Verma, 2018).

Inventory management and demand forecasting are closely related because it is essential to comprehend demand to maintain optimal inventory levels. Demand forecasting is a set of methods and procedures used to estimate the future demand for a product or service so that producers can plan the quantity of raw materials to order and manufacture, and retailers can maintain optimal stock levels (Ali & Govindan, 2021). The rise in digitalization has led to the production of vast quantities of data. The pattern analysis of data can aid in predicting future sales. Demand forecasting is crucial in inventory management. This determines a firm's profitability and competitive position. Predicting sales is the most important factor in determining retailers' inventory levels and order quantities (Kazancoglu et al., 2021; Kumar & Singh, 2021).

The term Fast-Moving Consumer Goods (FMCG) refers to products that are inexpensive and quickly sold. Because consumer purchasing behavior is highly erratic, it is challenging to predict future sales in this industry (Fosso Wamba et al., 2022; Grover et al., 2022). The FMCG market is the fourth-largest market segment in India, with primary products in three categories: household and personal care, healthcare, and food and beverages (FMCG). In India, revenues have increased by 24.4% over the past decade. The rural market for this sector is anticipated to grow at a compound annual growth rate (CAGR) of 14.6% and reach $220 billion by 2025. Because perishable products have a short shelf life, they are classified as FMCG. They require specialized transportation, are extremely sensitive to temperature changes, and require special maintenance; therefore, their inventory management becomes more complicated (Ali & Govindan, 2021).

The global wine market comprises three categories: still, sparkling, and fortified wine. It is the least widely consumed alcoholic beverage in the world. In 2019, wine accounted for 23% of the alcoholic beverage market and generated worldwide revenue of USD 378.3 billion. From 2012 to 2025, wine income is projected to increase at a CAGR of 5.6%, while sales volume is projected at a CAGR of 0.8% (Elluru et al., 2019; Katsaliaki et al., 2021). As a result of the COVID-19 pandemic, the estimated wine production decreased by 14.2%. In 2019, wine was predicted to generate the highest revenue among all wine categories, accounting for 85% of the wine segment's revenue. In 2020, it was found that the United States, France, and Italy consumed the most wine, while Italy, France, Spain, and the United States were identified as the largest wine-producing countries (Zhu, Krikke, Kumar, & Singh, 2021; Kazancoglu et al., 2021).

Owing to price-conscious consumers and a previous ban, wine consumption is lower in India than in other countries. While India has 123,000 acres of vineyards (as of December 2018), only 1–2% are utilized for wine production. The primary wine-producing regions are Nashik, Hampi Hills, Pune, Bijapur, Northern Karnataka, and Bangalore, with Nashik being India's wine capital (Kazancoglu et al., 2021). Between 2012 and 2013, the wine wholesale index increased by nearly 18%. In 2014, the quantity of imported wine steadily increased. In 2018, India imported approximately 5.2 million liters of wine, worth approximately $27.5 million. Companies in India may decrease their reliance on imports by improving their understanding of the market and using that knowledge in demand estimation to build resilient domestic supply chains (Zhu, Krikke, Kumar, & Singh, 2021).

The literature review identifies the following knowledge gaps: (a) Existing ML models lack explainability (Huber et al., 2020; Ali et al., 2009). Incorporating explainable AI (XAI) enhances transparency and trust, aligning it with responsible AI in operations research. (b) Most studies used single-stage models without segmentation (Huber et al., 2020; Tonbul, 2019). A two-stage approach can improve accuracy and provide granular insights. (c) Previous studies (Ali et al., 2009; Bala, 2012) used limited attributes for customer segmentation. It

is recommended that all available attributes are employed to enhance transparency and effectiveness. (d) Many studies focused on groceries and perishable goods (Ali et al., 2009; Bala, 2012). Few studies address the unique consumption patterns of the wine industry, emphasizing context-specific models and transparency. (e) Limited research has been conducted on utilizing sophisticated ML methods (SVM, VAR, and LSTM), and XAI forms a holistic platform that improves forecasting accuracy while maintaining ethical AI practices. To address these gaps, our study contributes to responsible AI in operations research by proposing a clear, explicable, and efficient framework for demand prediction in the perishable product supply chain, with a focus on the wine industry.

The major research questions are as follows: (1) Can transparent machine learning (ML) techniques be employed for customer-centric demand forecasting in perishable product supply chains? (2) Does ML enhance forecasting accuracy compared with traditional techniques?

To answer the abovementioned research questions, we set the following research objectives (ROs):

RO1: Hierarchical clustering is performed based on customer characteristics and XAI techniques are applied to identify the characteristics that distinguish customer segments.

RO2: Estimate wine consumption using machine-learning techniques for each identified segment.

RO3: Compare the efficacy of machine-learning techniques with conventional forecasting methods.

The remainder of this paper is organized as follows. Section 2 covers the winemaking process and the various AI and ML applications for inventory management and XAI. This section highlights the research gaps identified in the field and the contributions made as part of our research. The problem we are attempting to solve is discussed in Sect. 3. Section 4 describes the proposed solution framework for resolving the demand forecasting problem and uses realistic data to forecast the demand for perishable products. The outcomes of the proposed methodology are discussed in Sect. 5. Section 6 discusses the managerial implications of the proposed digital transformation method. Section 7 covers limitations, future research directions, and conclusions.

2 Literature review

A company's inventory management policy can either increase profits or result in losses. This essential task involves several crucial considerations, particularly concerning perishable goods. As the life expectancy of items decreases, they become more fragile, increasing challenges in their management. This study employed data from the wine industry. This section elaborately describes the winemaking process and concerns about perishable products.

2.1 Studies on perishable supply chains

Perishable supply chains refer to transporting and storing goods with limited shelf lives, such as fresh produce and dairy products. A study conducted by Cornell University offers insights into reducing food loss and waste in perishable supply chains (Tata-Cornell Institute, 2023). The study highlights that access to fresh fruits and vegetables is limited in developing countries such as India due to food loss and waste, among other factors. Another study examined supply chain design strategies for fresh produce, specifically using melons and sweet corn as examples (Blackburn & Scudder, 2009). This study explores the challenges

and opportunities associated with transporting and storing perishable products and provides recommendations for improving supply chain efficiency.

A recent study posited that competition among supply chain networks for perishable products has grown more intense because of the COVID-19 pandemic (Shafiee et al., 2022). This study provides critical insights for managers and decision-makers to mitigate the risks associated with perishable product supply chain networks in the current climate. The search results revealed recent studies on perishable supply chains. One study investigated the impact of the COVID-19 pandemic on perishable food supply chains across the globe and aimed to enhance their resilience (Shanker et al., 2022). Another study scrutinized the factors that influence perishable goods supply chains and emphasized the significance of effective time management in transporting perishable goods (Sharma et al., 2022). Moreover, a recent study in the United Kingdom examined the consequences of supply chain traceability and visibility on supply chain performance, sustainable practices, and velocity (Perishable Goods Supply Chain, 2023).

Overall, these studies highlight the importance of efficient supply chain management for perishable products to minimize food waste, increase access to fresh produce, and prevent disruptions to supply chains during unforeseen events such as pandemics. Our research examines supply chain design strategies for fresh produce, specifically using grapes and utilizing them to produce wine as an example of a perishable product. Wine is a grape-based, alcoholic beverage. There are numerous varieties of wine, most of which are made from grapes. Various types of wine are based on the grapes used, the environment in which they are grown, and the winemaking process. Three types of wine are red, white, and rose (Gupta et al., 2022).

The wine supply chain comprises four stages: harvest, manufacturing, packaging, and distribution. Researchers have studied optimizing these stages to fulfill wine orders at the lowest possible cost (Basso et al., 2023). Material shortages and supply chain challenges have affected the wine industry. Wineries and wine brands have reported shortages in packaging, production materials, and logistics headaches when trying to get their products to the market. This has led to increased costs and delays in the supply chain (Forbes, 2023). According to supply chain analysts, the schedule reliability of global ocean liners hit an all-time low in August 2021, which has caused shipping delays in the wine industry (Winemag, 2021). Overall, recent research on wine supply chains has focused on optimizing the different stages of the supply chain and addressing the challenges related to inventory management, material shortages, and shipping delays. These studies are important to ensure that the wine industry continues to operate efficiently and effectively.

2.2 Resilient inventory management of perishable product supply chains

Inventory management of perishable goods is complex because of their deterioration or spoilage, which can cause health problems if consumed later. Fruits and vegetables, meat, poultry, and other foods require proper storage to prevent the growth of bacteria, which can cause spoilage (Zhu & Krikke, 2020; Kumar & Singh, 2021; Kazancoglu et al., 2021). They must be refrigerated at temperatures as low as 40 °F or frozen at 0 °F or lower, necessitating expensive equipment, substantial initial investment, and substantial maintenance expenses (Yavari & Zaker, 2019, 2020; Kumar et al., 2021).

The transportation of perishable goods is challenging. For instance, once fruits and vegetables are harvested, they ripen by consuming atmospheric oxygen and quickly perish. The respiration rate must be controlled without causing harm to maintain the proper texture,

appearance, and flavor of fruits and vegetables (Zhang et al., 2021). It is essential to find appropriate packaging based on the type of fruit or vegetable and to determine the composition of respiratory gases to keep fruits and vegetables fresh until they reach the end-user or a factory where they will be further processed (Yavari & Zaker, 2020).

The COVID-19 pandemic has significantly disrupted the operation of perishable supply chains and exposed the limitations of demand forecasting. Specifically, the following issues have arisen: (a) Disruptions: The imposition of lockdowns and travel restrictions has resulted in a shortage of labor, transportation delays, and border closures, which have negatively affected the transportation of perishable goods. This, in turn, leads to commodity spoilage and inventory shortages. (b) Fluctuating Demand: Localized incidents of panic buying and stockpiling have caused unpredictable spikes in demand, followed by a sharp decline. Conventional forecasting models are inadequate for handling erratic consumer behavior. (c) Difficulties in Forecasting: The sudden and unforeseeable nature of COVID-19 presents challenges in accurately forecasting the demand. Conventional models have not yet been designed to address such significant disruptions (Masudin et al., 2021; Kumar et al., 2022; Ghafour et al., 2023).

2.3 AI applications for resilient perishable supply chains

Researchers now have vast amounts of data that allow them to comprehend market dynamics and develop more accurate methods for predicting demand and enhancing the supply chain value (Kumar et al., 2021). Owing to the vast amount of data currently available, AI and ML techniques can be used to improve supply chains. Some of AI's most crucial supply chain outcomes are demand forecasting, risk and resilience management, transportation, supplier selection, and inventory management (Riah et al., 2021). Table 1 summarizes some research on inventory management and demand forecasting.

Inventory management is divided into three categories: (1) raw materials, (2) work in progress, and (3) finished goods ready for shipment. Given the number of actors involved in each stage, it is essential to establish coordination and define appropriate supply chain policies (Gligor et al., 2019; Modgil et al., 2021; Zhang et al., 2021). This topic relies heavily on demand forecasting and has unrealized potential.

2.4 Explainable AI

There is a growing concern that, as the underlying AI models have become increasingly complex and powerful, they have also become a "black box" lacking comprehensibility (Meske et al., 2022). A comprehensible model is necessary to deem AI models fair and trustworthy and foster their usage (Abdul et al., 2018). The field of XAI has undergone significant development over the years. Arrieta et al. (2020) comprehensively reviewed the XAI and discussed related concepts, taxonomies, opportunities, and challenges. Similarly, Adadi and Berrada (2018) conducted a comprehensive XAI survey to explore various techniques and methods. Here, we summarize the need for XAI.

AI techniques have evolved from straightforward and interpretable methods, such as linear models, to increasingly intricate and sophisticated approaches, such as ensembles and deep learning. This progression led to an increase in the complexity and opacity of the models. Opaque models such as DL have consistently demonstrated better performance. Hence, there has been a continuous effort to enhance the interpretability of these opaque models. This pursuit of XAI aims to ensure that the model's inner workings are clear to decision-makers.

Table 1 AI Applications in inventory management and demand forecasting

Reference	AI algorithm / Technique	Sector/Industry of application	Contributions
Ali et al. (2009)	ML	Grocery	ML techniques dominated traditional statistics, outperforming benchmark MAE by up to 28.4% in some instances
Bala (2012)	ANN	Retail	Clustering-based seasonal ARIMA with predictors had the lowest MAPE and NSME. The improvement over the moving average model is around 60%
Lolli et al. (2018)	ML	-	SVM outperformed DNN in terms of classification accuracy
Praveen et al. (2019)	ANN	Retail	ANN models improved forecasting accuracy by 2–3%
Tanizaki et al. (2019)	ML, statistical analysis	Restaurant	Forecast accuracy did not differ much between Bayesian Linear Regression, Boosted Decision Tree Regression, Decision Forest Regression, and Stepwise methods
Weber et al. (2019)	AI	Retail	Describe the applications of AI in retailing—managing goods, ordering goods, serving customers, transporting goods, handing out goods, and financial accounting
Huber et al. (2020)	ML	Retail/ Bakery	Several machine learning methods were compared, showing that LSTMs outperform other ML techniques by a factor between 1.0 and 1.4 for multistep forecasting
Tian et al. (2021)	Markov-combined method	Retail	Markov combined methods outperformed other intermittent forecasting methods by a minimum of 25% in MAE
Spiliotis et al. (2022)	ML	Retail	While statistical models depicted inferior performance, most machine learning models depicted superior performance (up to 5.0%) in RMSSE compared to the Croston method
Feizabadi (2022)	ANN	Steel	ML-based models yielded, on average, a 5% improvement in forecast accuracy (1-MAPE)
Pereira and Cerqueira (2022)	ML	Hotel	ML models reduced RMSE by 54% and 45% for 1-day and 14-day horizons
Viverit et al. (2023)	ML	Hotel	Cluster-based forecasting depicts higher accuracy compared to non-cluster-based forecasting

Table 1 (continued)

Reference	AI algorithm / Technique	Sector/Industry of application	Contributions
Peláez-Rodríguez et al. (2024)	ANN	Emergency department visits	The DL-based approach yielded improvements in MAE of at least 8% and 3% in two datasets predicting emergency department visits
This study	ML	FMCG	Provides a method to improve supply chain values through better demand forecasting through customer segmentation

For instance, in the context of this study, decision-makers may seek to comprehend the factors that distinguish clusters (segments). Without this understanding, they may hesitate to apply forecasting techniques at the segment level. We chose XAI because it allows us to use black-box models for demand forecasting, which are superior to white-box models in performance.

Dwivedi et al. (2022) compared and contrasted XAI feature-based XAI techniques: feature importance, partial dependence plots, individual conditional expectation, global surrogate, LIME, and Shapley values. These techniques differ in the extent to which they are globally or locally interpretable, model-specific, or model-agnostic. Global interpretable techniques analyze the overall decision-making process of a model, whereas local interpretable techniques analyze the decision-making process at an instance level. Model-specific interpretation techniques can only work with specific ML or CL algorithms; model-agnostic techniques do not have this disadvantage. Among the aforementioned feature-based XAI techniques, Shapley values have the advantage of being globally interpretable and model-agnostic. We explain the difference between the Shapley values and LIME in detail in Sect. 4.4.

A few applications of XAI are as follows. Abdollahi and Nasraoui (2018) discussed the importance of transparency in fair machine learning, specifically in the context of explainable recommendation systems. Different approaches have been proposed to achieve explainability in AI models. Angelov and Soares (2020) introduced the concept of explainable deep neural networks (xDNN), which aimed to provide interpretable representations in deep learning models. Al-Shedivat et al. (2020) presented contextual explanation networks, which aimed to capture the dependencies among explanations. Samek et al. (2019) explored the interpretability of Long Short-Term Memory (LSTM) models, specifically focusing on explaining and interpreting their behavior. These studies highlight the diversity of methods employed to achieve explainability in AI.

AI techniques have gained significance in the demand forecasting of perishable product supply chains. A literature review by Mediavilla et al. (2022) analyzed various artificial intelligence methods for demand forecasting in supply chain management, including machine learning and deep learning. Although not specific to perishable product supply chains, this review explains how AI techniques can be applied to demand forecasting in various domains.

While the literature on the application of ML to solve supply chain problems is abundant, there is a paucity of research concerning the incorporation of XAI. Further research is required to bridge this gap and address the unique challenges of demand forecasting for perishable products. We address this concern by adopting XAI techniques in this study.

2.5 Research gaps

Huber et al. (2020) investigated the bakery chain case and projected daily demand for various product categories in each store. In this study, Artificial Neural Networks (ANN) and decision trees were among the models employed. The results demonstrate that Machine Learning (ML) methods are suitable for forecasting because of their superior performance and capacity to handle larger-scale demands. It was found that classification techniques surpassed regression techniques in forecasting, which is a noteworthy observation. According to Ali et al. (2009), the perishable nature of groceries makes demand forecasting particularly challenging, necessitating advanced models. This study examines demand forecasting during sales in a European supermarket. The study revealed that pooling data enhanced the performance of the model. An interesting finding was that a simple time-series method worked well in the absence of promotions; however, a more complex method, such as regression trees

with appropriate features, is required for accurate predictions when promotions are available (Ivanov, 2021; Liu & Lee, 2018; Mishra et al., 2021).

Following these early studies, ML techniques are generally superior to traditional approaches because of their ability to adapt to changing patterns, model complex and nonlinear associations, versatility in handling structured and unstructured data, high-dimensional data, and feature engineering abilities. Hence, we chose ML techniques for the forecasting.

Our research employs machine learning techniques but distinguishes itself from existing ML-based forecasting studies in several ways. Most prior studies have not adopted a two-stage approach or incorporated XAI. For instance, Ali et al. (2009) utilized regression trees for demand forecasting, whereas Praveen et al. (2019) applied artificial neural networks (ANN). Lolli et al. (2018) employed support vector machines (SVMs), and Huber et al. (2020) and Tanizaki (2019) explored multiple ML techniques. Tian et al. (2021) implemented Markov Chains. However, none of these studies considered customer-centric segmentation, as we have done. Bala (2012) used segmentation followed by demand forecasting. However, unlike Bala (2012), who used five attributes based on rule-based classification, our approach is more flexible and uses all available attributes to create segments. Our approach is lenient and enhances the likelihood of identifying naturally occurring customer segments.

Following clustering, we also increased comprehensibility and confidence by using XAI to determine the characteristics that differentiate the segments. Furthermore, we also considered forecasting techniques such as the support vector method (SVM), vector autoregressive (VAR), and long short-term memory networks (LSTM). Another difference is that the context of the product in our case is wine, which varies in consumption patterns from the grocery products considered in Bala (2012). The contributions of this study are as follows.

- This study proposes a solution framework for perishable supply chains that segments customers based on their characteristics and uses transparent and responsible artificial intelligence techniques for demand forecasting.
- We employed k-means, hierarchical clustering, and XAI for segmentation and modeling and compared several machine learning techniques for demand forecasting.
- This study demonstrated the use of XAI to enhance transparency in machine learning and improve forecasting accuracy, which can enhance the resilience of supply chains for perishable products.
- The main implication of this study is the demonstration of XAI in enhancing transparency in ML, and an improved method for reducing forecasting errors in practice, which can enhance the resilience of supply chains for perishable products.

3 Problem statement

Perishable product supply chains are characterized by distinct challenges compared to non-perishable products, necessitating stricter management practices throughout the process (Bharti, 202). These challenges can be summarized as follows. (a) Perishability: The limited shelf life of perishable products mandates precise control over storage, transportation, and handling to minimize spoilage. This requires careful planning, coordination, and risk-management strategies (Osman et al., 2023). (b) Temperature sensitivity: Many perishables require specific temperature ranges during storage and transportation to maintain quality. Deviations can result in rapid spoilage and significant waste generation (Vrat et al., 2018). (c) Demand variability: The demand for fresh produce can fluctuate significantly depending on seasonality, weather patterns, and consumer preferences. This unpredictability makes it

challenging to optimize inventory management compared with non-perishables with more stable demand (Ali et al., 2021). (d) Quality Deterioration: Perishable foods are sensitive to time and temperature, and quality deterioration increases with increased shipping times. Demand unpredictability exacerbates this challenge (Taparia et al. 2020).

Figure 1 depicts a wine supply chain that was simplified to reduce complexity. It is assumed that retailers can purchase directly from manufacturers without the need for distributors or other intermediaries and that a few steps are combined into a single stage (Fosso Wamba et al., 2022; Grover et al., 2022; Gupta et al., 2022).

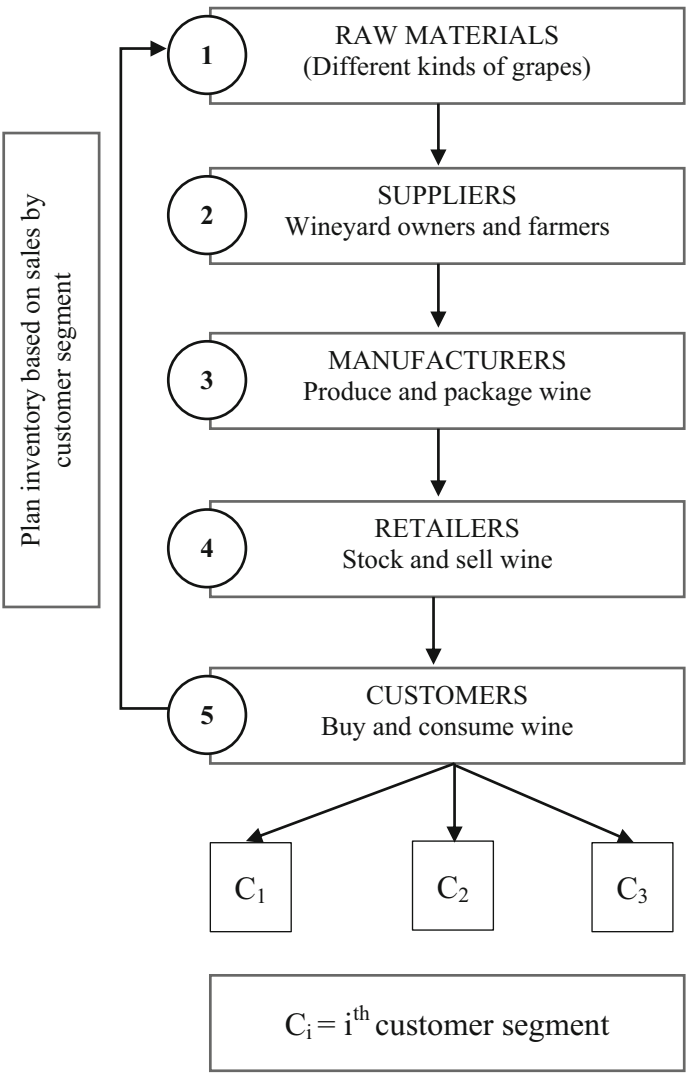


Fig. 1 Traditional demand forecasting

Inventory management is a crucial aspect of the perishable supply chain; stockouts reduce customer satisfaction, which results in low revenues, whereas overstocking increases inventory holding costs, which increases inventory costs. Inventory management is closely related to the demand for products or services. As such, it depends on the precision of the demand forecasting (Ali & Govindan, 2021). Optimizing inventory levels from beginning to end is essential for increasing the value of the entire supply chain. Owing to stockouts and the lead time associated with stock replenishment, retailers frequently struggle to provide customers with desired products at the right time, negatively affecting customer satisfaction (Ivanov, 2021; Liu & Lee, 2018; Mishra et al., 2021).

Although India produces large quantities of grapes annually, its wine industry is still in its infancy. Because wine's primary raw material is grapes, a perishable product, inventory management becomes even more challenging. As wine companies are new, if they invest heavily in inventory, they will lose money, and the industry will perish. Therefore, it is crucial for the company to understand the demand for wine based on the size of its customer base and to produce sufficient quantities each year to meet those demands. As most of the wine consumed in India is imported, the industry must embrace technology to achieve revolutionary growth.

In the conventional method, sales from previous years are used to predict sales for the next year, increase or decrease production, and plan for raw materials. Unaddressed is the fact that sales are dependent on customers and that there are customer segments whose characteristics may affect product sales, thereby influencing the inventory levels that the company must maintain. The raw materials used to produce wine are various varieties of grapes, and the suppliers are farmers and vineyard owners. After harvesting and producing wine, manufacturers collect the grapes and distribute them to retailers.

4 Solution approach

This section describes the procedure and steps for implementing the model. Figure 2 summarizes the methods used in this study. Initially, extensive research was conducted to identify the issue of inventory management in perishable products. In this dataset, our primary raw material is grapes, and we are attempting to forecast wine demand. The quantity of wine produced is determined by the quantity sold. Figure 1 illustrates the relationship between the demand for raw materials and consumer purchases. We then test the robustness of the generated clusters by building a surrogate model and describe the clusters by gathering insights from the surrogate model via XAI techniques. Forecasting techniques are then used to predict the demand for each cluster. The proposed method aligns with standard frameworks such as CRISP-DM, which suggests sequence – business understanding, data understanding, data preparation, model development, model evaluation, and deployment. In our approach, we developed and validated two models: clustering and forecasting, as shown in Fig. 2 (Wang et al., 2019).

4.1 Data collection

Data were gathered from Kaggle and evaluated to determine their suitability for use in this study (Kaggle). The dataset contains information on the customers of X. It contains 2241 observations and 29 attributes, each representing a characteristic of a wine purchaser. Table 2 describes the attributes, and Table 3 displays the descriptive statistics of the selected attributes.

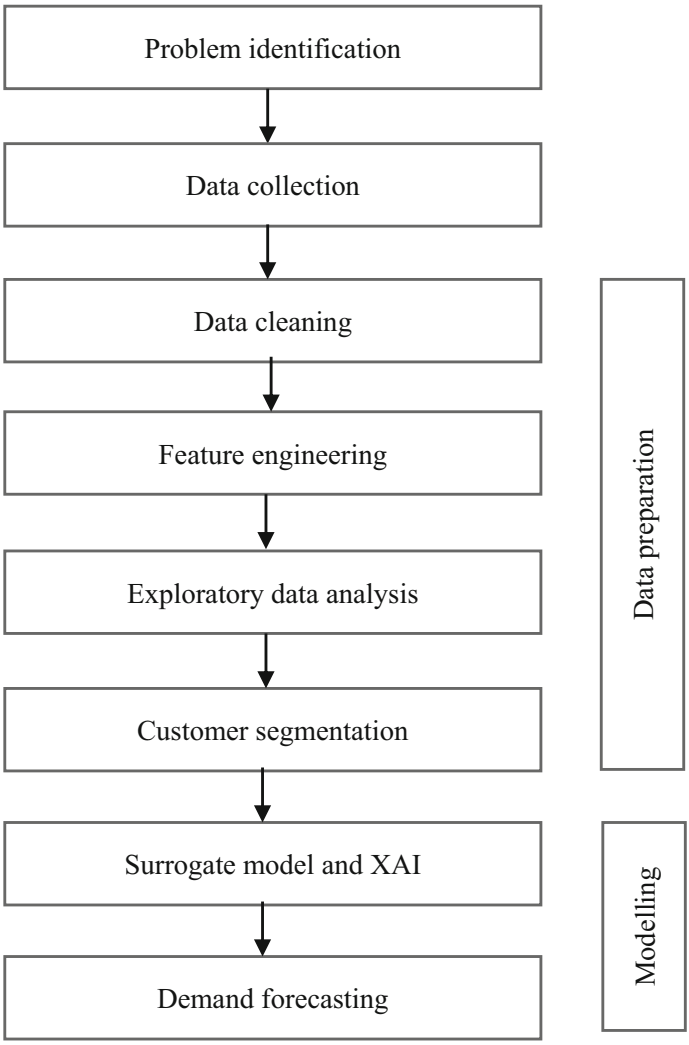


Fig. 2 Segmentation and estimation framework

This dataset is helpful for supervised and unsupervised machine learning tasks. A limitation of the dataset is that country or location information was unavailable. Hence, it may not be suitable for analyzing location-specific conclusions. This dataset can generally be used for modeling and validating regression, classification, clustering, forecasting models, and testing new approaches for such problems. Jupyter Notebook and Tableau were utilized, and the code was written using Python. The ML steps were executed on a 64-bit machine with 8 GB of RAM and Windows 10 operating system.

Table 2 Description of variables in the dataset

Name of the column	Description
ID	Customer identity number
Kidhome	Number of kids at home
Education	Education of customer
Year_Birth	Birth year of customer
Marital_Status	Marital status of the customer
Income	Income of customers per annum
MntFruits	Amount spent on fruits in last two years
Teenhome	Number of teenagers at home
Dt_Customer	Date of last purchase
Recency	Number of days since last purchase
MntWines	Amount spent on wine for two years
MntSweetProducts	Amount spent on sweets in last two years
MntMeatProducts	Amount spent on meat in last two years
MntFishProducts	Amount spent on fish in last two years
MntGoldProds	Amount spent on gold in last two years
NumDealsPurchases	Number of purchases which got discount
NumWebPurchases	Number of purchases made through the company's website
NumCatalogPurchases	Number of purchases using catalog
NumStorePurchases	Number of purchases made in the store
NumWebVisitsMonth	Number of times a customer visited the company website
AcceptedCmp1	Customer utilized campaign-1 offer or not—1 if accepted, 0 if rejected
AcceptedCmp2	Customer utilized campaign-2 offer or not—1 if accepted, 0 if rejected
AcceptedCmp3	Customer utilized campaign-3 offer or not—1 if accepted, 0 if rejected
AcceptedCmp4	Customer utilized campaign-4 offer or not—1 if accepted, 0 if rejected
AcceptedCmp5	Customer utilized campaign-5 offer or not—1 if accepted, 0 if rejected
Complain	Customer complaints in last two years—1 if complained, 0 otherwise
Z_CostContact	It contains 3 in all the rows, so it is not used in model
Z_Revenue	It contains 11 in all the rows, so it is not used in model
Response	Customer accepted offer in last campaign—1 if accepted, 0 if rejected

4.2 Data preprocessing

Features are engineered to make the data conducive to modeling through feature engineering. The data were cleaned by removing missing values and outliers. Furthermore, new features were constructed from existing features, and categorical variables were encoded using one-hot encoding. Because the solution approach involves clustering, the existing features are standardized. The feature engineering approach adopts the best practices followed in machine learning (Wirth & Hipp, 2000).

Table 3 Descriptive statistics of data

Column	Min	Max	Mean	STDEV
Income	1730	666,666	52,247.25	25,173.08
Kidhome	0	2	0.44	0.54
Teenhome	0	2	0.51	0.54
Recency	0	99	49.11	28.96
MntWines	0	1493	303.94	336.60
MntFruits	0	199	26.30	39.77
MntMeatProducts	0	1725	166.95	225.72
MntFishProducts	0	259	37.53	54.63
MntSweetProducts	0	263	27.06	41.28
MntGoldProds	0	362	44.02	52.17
NumDealsPurchases	0	15	2.33	1.93
NumWebPurchases	0	27	4.08	2.78
NumCatalogPurchases	0	28	2.66	2.92
NumStorePurchases	0	13	5.79	3.25
NumWebVisitsMonth	0	20	5.32	2.43
Response	0	1	0.15	0.36
Complain	0	1	0.01	0.10

4.2.1 Data cleaning

Data containing missing values, outliers, or extreme values tend to render the results unreliable. Recommended measures, as per Hair et al. (2016), were adopted for this purpose. For example, we observed the ages of the children for each observation. If the individual has one or no children, the information about the other child is marked as Not Available (NA) or left blank. As the missing values represented less than 10% of the data, these rows were eliminated rather than having their values imputed. Attributes "Age" and "Income" were found to contain outliers, and the rows were removed.

4.2.2 Feature construction

This is the process of developing features conducive to model building. Table 4 provides a list of features developed using the feature construction method based on the existing features.

4.2.3 Feature elimination

Eliminating features helps reduce model complexity. Attributes that did not contain meaningful information or were rendered redundant after the feature construction were eliminated, as shown in Table 5.

4.2.4 Exploratory data analysis (EDA)

An exploratory analysis of the data revealed the following, as shown in Fig. 3. Most of the

Table 4 Feature engineering

Feature	Description
Age	Denotes the age of the person Age = 2021-Year_Birth
MntTotal	Denotes total amount spent on different things except wine for two years MntTotal = MntFruits + MntMeatProducts + MntFishProducts + MntSweetProducts + MntGoldProds
Living_With	Tells if someone is alone or has a partner
Education	Denotes education of the person whether someone is Undergraduate, Graduate, Postgraduate
Children	Denotes total number of children Children = Kidhome + Teenhome
Family_Size	Denotes size of family Family_Size = Number of children + 1 if Marital status is Alone Family_Size = Number of children + 2 if Marital status is Partner
Is_Parent	Denotes if someone has children or not Is_Parent = 1 if the customer has children, 0 otherwise
Customer_for	Denotes difference in days from the last purchase recorded to the day on which the customer made the purchase
Numpurchases	Obtained by summing up purchases made across different modes

customers were graduates (~ 50%) and had partners (65%). Approximately 81% of customers have one or more children. Most customers have annual incomes of between $40 and $80 thousand. While spending on wine consumption varied from $0—$1500, 50% of customers spent less than $300. Graduates and postgraduates spend more than an average on wines. There is little difference between customers based on marital status. Spending on wines was positively correlated with income and reduced with an increase in family size.

4.2.5 Encoding categorical variable and standardization

The dataset contained two categorical variables: "Living_With" and "Education." We used dummy coding, which resulted in new attributes. The attributes take a value of zero or one, indicating whether they represent something. Standardization is one of the techniques used to transform each attribute into an attribute with a zero mean and one standard deviation. This is useful when the attributes vary by an order of magnitude. The following columns were scaled using standardization: income, recency, MntWines, NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisits, Month, Age, MntTotal, Children, Family Size, and Customer For.

4.3 Customer segmentation

Customer segmentation identifies homogenous customer groups based on the shared characteristics. We used k-means clustering and agglomerative hierarchical clustering to create customer segments (Bala, 2012; Viverit et al., 2023). These two methods allowed us to evaluate the robustness of the cluster (segments). We then build a surrogate model using a classification technique to ensure the clusters are robust and then apply XAI methods to describe the resulting clusters. These clusters, which were tested and described, served as the basis for demand forecasting.

Table 5 Dimensionality reduction

Removed column	Reason
ID	This column denotes only the unique ID for each transaction
Marital_Status	Replaced by feature: Living_with
Dt_Customer	Replaced by feature: Customer_for
Z_CostContact	All the rows in this column contain a unique value
Z_Revenue	All the rows in this column contain a unique value
Year_Birth	The age feature explains this feature
Kidhome, Teenhome	These features are explained in children's feature
MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds,	The Mnt_Total feature replaces these features
NumDealsPurchases, NumWebPurchases	
NumCatalogPurchases, NumStorePurchases	The Numpurchases feature replaces these features
AcceptedCmp3, AcceptedCmp4, AcceptedCmp5, AcceptedCmp1	
AcceptedCmp2	This may be more specific to this case, so removed

4.4 XAI

Section 2.4 describes the literature that highlights the need for transparency and interpretability in machine learning (Abdollahi & Nasraoui, 2018; Abdul et al., 2018; Meske et al., 2022). In this study, clustering techniques were used to determine customer segments. However, we do not know (a) whether the segments are robust and (b) their distinguishing features. This is because, while clustering techniques are not entirely black-box, they are not as amenable to interpretation as white-box-supervised models. The role of XAI in this study is to address these issues by building interpretable surrogate models. Surrogate models are usually white-box techniques such as linear regression, logistic regression, classification, and decision trees. We choose an ensemble method, namely random forest, as it performs better as a surrogate model and is interpretable using several XAI techniques. The contribution of XAI is that it lends credibility to this study by rendering the ML approach transparent.

The XAI process was operationalized in two steps. The first step involves building and evaluating a random-forest-based surrogate classification model using the same features as the clustering model. The clustering process is deemed robust if the random forest model, as developed, shows high accuracy. The second step involves XAI methods, such as feature

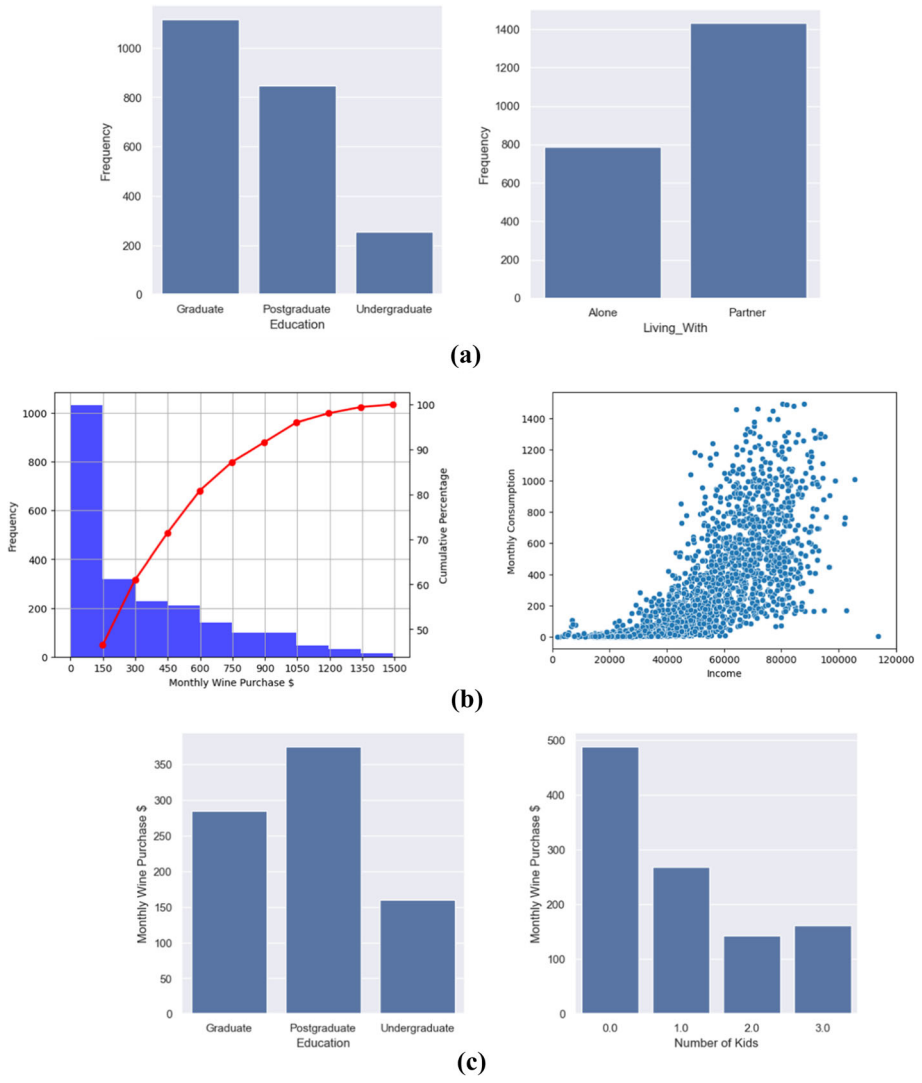


Fig. 3 Exploratory analysis of wine consumption data **a** Histograms of customers by education and marital status **b** Histogram of monthly spending and income vs monthly spending on wines **c** Monthly spending on wines by education and family size

importance and Shapley values, to determine the features that contribute the most to discriminating the clusters. XAI offers several model-agnostic approaches for interpreting black-box models such as LIME and SHAP (Gramegna & Giudici, 2021). Locally interpretable Model Agnostic Explanation (LIME). LIME aims to create a model to explain the prediction for an observation using data points similar to the explained observation while minimizing the loss in making such predictions (Ribeiro et al., 2016). On the other hand, the Shapley Additive Explanation (SHAP) returns Shapley values that indicate the contribution of each predictor to each observation (Lundberg & Lee, 2017). The difference between the LIME and Shapley

values is that the Shapley value provides both local and global explanations, while LIME provides only a local explanation (Dwivedi et al., 2023). Global explanation means that Shapley values can provide insights into the importance of features across the entire dataset, unlike LIME, which can provide valid interpretations at an instance level. This helps us to understand the feature's importance and why an instance's feature values lead to a specific outcome. Unlike LIME, which is less formalized, SHAP has a strong foundation in cooperative game theory (Angelov et al. 2021). LIME interpretations are based on the perturbation of the input data and may be unreliable if such perturbations are not carefully chosen (Angelov et al., 2021).

Another method, namely the relative importance method, is specific to tree-based methods, and it ranks feature contributions by averaging their contribution to improving the homogeneity of nodes across several trees. We used three methods for global explanation: SHAPKernelExplainer (a model-agnostic method), SHAPTreeExplainer (a tree model-specific method), and Relative Importance in determining the features with the highest discriminator power. Using three different approaches for XAI helps us assess whether the top-ranking features appear consistently across all the approaches.

Please refer to Appendix A for further information on the models and their configurations. For a detailed description of the techniques employed in our study, please refer to Appendix B, which provides a comprehensive overview of the following techniques: hierarchical clustering, random forest, long short-term memory (LSTM), support vector regression (SVR), autoregressive models, vector autoregression, and feature importance using Shap.

4.5 Demand forecasting

The dataset consisted of time-series data, meaning that each data point was associated with the timestamp at which it was recorded. We added cluster labels identified using hierarchical clustering to the dataset. To apply forecasting techniques, the data must be reordered at regular intervals of time, such as daily, weekly, or monthly; however, our current dataset was not recorded at regular intervals, with data related to only 662 days indicating sales over three years from 2012 to 2014, despite having a total of 1096 days. As we forecast the demand for four clusters over a given number of years, we created three years of realistic data for each cluster, from January 1, 2012, to December 31, 2014, for 1096 days per cluster.

4.5.1 Data creation

To generate realistic data, we assumed that the amount spent on wine followed the same monthly trend. Therefore, we used a random number generator to determine the maximum and minimum values for each month and missing data. If we did not have any values for a given month in a given cluster, we used four adjacent months of data and the same method described in the preceding paragraph to generate the data. Once we had data for the four clusters, we calculated the total number of wines sold over three years and the total number of website visits and revenue. To determine the values of the other attributes, we calculated the average value for each attribute across all dates. This information was used to predict future wine sales. Table 6 presents the original and forecasted data.

Table 6 presents the original and forecasted data

Original Data						
Date	Income	MntWines	MntTotal	NumWebVisitsMonth	NumPurchases	Complain
08–01-2012	7500	7	64	9	12	0
08–01-2012	82,576	1206	704	1	19	0
08–01-2012	7500	5	95	9	10	0
08–01-2012	73,803	833	695	6	22	0
09–01-2012	94,871	169	909	7	18	0
10–01-2012	51,563	1166	84	8	19	0
Generated Data						
Date	Income	MntWines	MntTotal	NumWebVisitsMonth	NumPurchases	Age
01–01-2012	234,673.415	2160.425	923.010	25.256	65.799	49.937
02–01-2012	211,597.914	1306.570	1696.370	33.403	16.492	55.643
03–01-2012	239,429.180	2448.330	1068.422	33.124	13.544	55.403
04–01-2012	236,590.831	1227.921	833.472	32.245	13.755	52.213
05–01-2012	161,143.645	1754.867	1370.595	29.797	16.871	51.132
06–01-2012	171,626.623	1172.548	1202.909	29.738	16.302	45.299

4.5.2 Forecasting models development

We developed and compared several forecasting models and benchmarked them against a naive forecast. The forecasting models that we developed and tested included statistical, ML, and DL models, which align with the literature summarized in Table 1. Evaluating the accuracy of forecasts generated by various models is essential. To determine how well our model performs on unknown data, the dataset was divided into training and testing data, with the most recent 90 days of data used for testing and the remainder used for training the model. The models were built using training data. We used the Root Mean Square Error (RMSE) as a performance measure to determine model accuracy. The forecasting model with the lowest RMSE for the test data is the most accurate.

$$\text{RMSE} = \sqrt{\frac{1}{\text{n}}\sum_{\text{i=1}}^{\text{n}}(\text{Actual}_i - \text{Predicted}_i)^2} \tag{1}$$

where: Actual_i = actual value of ith observation, Predicted_i = predicted value of ith observation.

5 Results and analysis

This section presents the research findings and an in-depth analysis of the four sections.

5.1 k-means clustering

K-means clustering is a partitioning-based approach for clustering. This method generates and assigns data points to k clusters using the following steps:

Step 1: Starting with a given value of k
Step 2: Randomly identify k points as centroids.
Step 3: Each point is assigned to the centroid closest to it.
Step 4: Recalculating the centroids of k clusters.
Step 5: Repeat Steps 3 and 4 until the centroids do not change in successive iterations.

We applied k-means clustering to the data obtained after standardization. A scree plot was used to determine the optimal number of clusters. Using a scree plot, we determined the locations where we achieved the lowest within-cluster sum of squares (WCSS) with the fewest clusters. We found that the WCSS did not drop appreciably beyond these five clusters. We obtained four clusters using $k = 4$ and analyzed them based on the average age, average number of children, average education, average family size, average income, and having children. This process is repeated for $k = 5$. Table 7 shows a few characteristics of the four clusters identified using k-means.

5.2 Hierarchical clustering

Hierarchical clustering is the hierarchical category of clustering methods. There are two types of hierarchical clustering: divisive and agglomerative. We used the agglomerative clustering method using distance type "Euclidean" and linkage type "Ward" to derive four- and five-cluster solutions. Table 8 lists the characteristics of the four clusters identified using

Table 7 K-means, 4- cluster solution

Characteristics / Cluster name	Cluster 0	Cluster 1	Cluster 2	Cluster 3
Size	666	586	494	466
Average age	45.13	55.33	52.99	57
Average children	0.8	1.11	0.03	1.92
Education	51% graduates	51% graduates	53% graduates	46% graduates
	29% postgraduates	42% postgraduates	39% postgraduates	46% postgraduates
	20% undergraduates	7% undergraduates	8% undergraduates	8% undergraduates
Average family size	2.42	2.78	1.61	3.64
Average income	30,880	61,947	77,113	42,858
Having children	79% of them have children	97% of them have children	3% of them have children	100% of them have children
Living with	38% alone	33% alone	42% alone	28% alone
	62% living with a partner	67% living with a partner	58% living with a partner	72% living with a partner

Table 8 Hierarchical four-cluster solution

Characteristics / Cluster name	Cluster 0	Cluster 1	Cluster 2	Cluster 3
Size	675	593	480	464
Average age	45.27	55.14	52.85	57.3
Average children	0.77	1.16	0.01	1.91
Education	50% graduates	50% graduates	52% graduates	50% graduates
	30% postgraduates	43% postgraduates	40% postgraduates	42% postgraduates
	20% undergraduates	7% undergraduates	8% undergraduates	8% undergraduates
Average family size	2.41	2.81	1.61	3.59
Average income	31,330	62,245	77,509	42,392
Having children	77%	100%	1%	100%
Living with	36% alone	34% alone	40% alone	32% alone
	64% living with a partner	66% living with a partner	60% living with a partner	68% living with a partner

hierarchical clustering.

5.3 Selection and analysis of cluster solution

By examining the size of the clusters formed and the variation in characteristics across solutions, we identified that the characteristics of clusters in the k-means four-cluster solution and hierarchical four-clustering solution were more similar than their five-cluster counterparts. The four customer segments, Segment "0," Segment "1," Segment "2," and Segment "3" correspond to the four clusters in Table 7. We retained the four segments from hierarchical clustering. As described earlier, we first built a surrogate model to validate the robustness of the clustering. In the second step, we employ XAI methods such as feature importance and Shapley values to determine the features that contribute the most to discriminating the clusters. The features identified would help us describe the clusters.

We built a random forest classifier in the first step. The dependent variable was the cluster label and the independent variables were the same as those used in the cluster analysis. We used 70% of the data as the training data and 30% as the testing data. Upon evaluation using test data, we found that the classification model had an accuracy of 96%. As shown in Table 9, the precision, recall, and f1-scores for all four clusters (segments) exceeded 92%, thereby validating that the clustering task produced highly different clusters.

As the clustering process was deemed trustworthy, we extracted the features that discriminated the clusters using relative importance and Shapley methods. Figure 4 shows the most important features gauged using the relative-importance method. The higher the relative importance score of a feature, the higher its discriminatory power. Likewise, Fig. 5 and Fig. 6 rank the features based on their contribution using SHAP KernelExplainer and SHAP TreeExplainer. A comparison of Figs. 4, '5, ' and 6 shows 80% commonality among the top five features and 100% commonality among the top '10' features. The commonality among the methods suggests the procedures adopted and, eventually, the resulting clustering solution is robust. Hence, the subsequent classification model can discriminate the clusters well. Table 10 presents the cluster (segment) summaries of the top 10' features based on the Shapley values.

These segments were distinct from each other. What distinguishes them? This insight was obtained from Shapley values. In Figs. 5 and 6, the higher a feature scored on the Shapley value, the more impactful it is in distinguishing segments. The insight from Figs. 5 & 6 is that the features, namely, 'Children, ' 'MntWines,' 'Income, ' 'Is_Parent,' 'NumPurchases,' 'Family_Size,' 'MntTotal,' 'Age, ' 'NumWebVisitsMonth,' and 'Customer_For' are the top 10 distinguishing set of features among all features. The results are critical for the study because we eventually describe the segments using these features, and forecasting is performed at the segment level.

Table 9 Classification Metrics

Cluster	Precision	Recall	F1-score	Support (Test **N**)
Cluster 0	0.96	0.97	0.97	197
Cluster 1	0.94	0.92	0.93	183
Cluster 2	0.99	0.99	0.99	151
Cluster 2	0.93	0.95	0.94	133

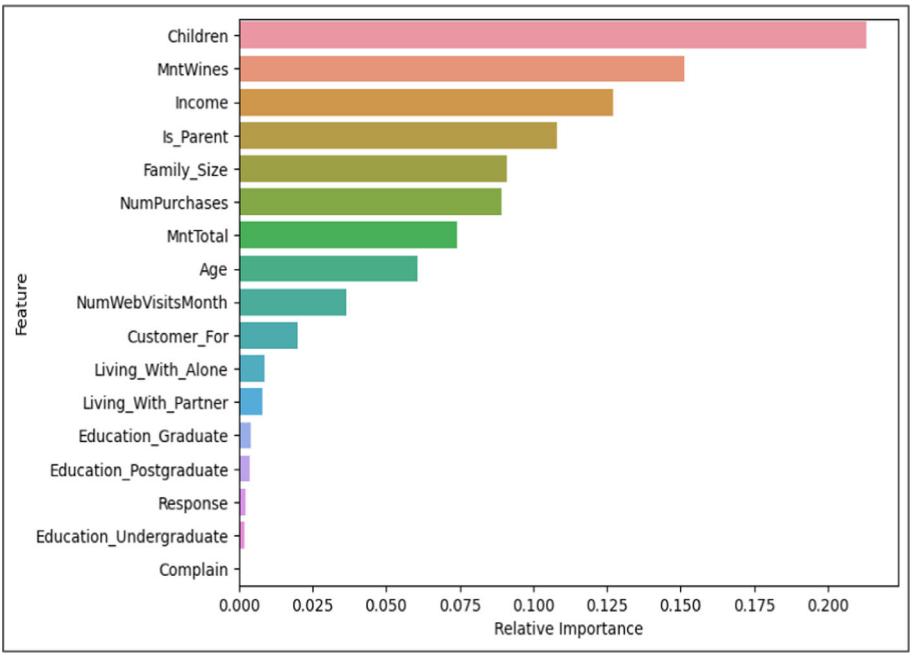


Fig. 4 Feature rank using the relative importance method

The use of these features to describe the segment increases interpretability. While Shapley values explain a solution or model, interpretability refers to how well the solution is amenable to human interpretation. As evident from Table 10, Segment (cluster) "3" ranks highest in terms of the number of children and family size, Segment "2" has the highest income, overall monthly spend, and monthly spend on wines, Segment "1" and "2" make frequent wine purchases, Segment "3" has the largest family size, Segment "0" is the youngest segment and is most active on the website and Segment "1" contains long-term patrons. Here, we describe the individual segments.

Segment "0": This is the largest segment, representing 31% of the market with the lowest income (31 K), lowest monthly spending on wines (46.9), and does not make frequent purchases (8.4). They are relatively young consumers (45 years old) with the second lowest monthly overall spending (84.03) on foods other than wine.

Segment "1" is the second-largest segment, representing 27% of the market with the second-highest income (62 K). They spend almost 11 × times (512.5) on wines, with a frequency of 2.5 × times (21.6) compared to Segment "0." They are relatively older consumers (55 years), have a partner and more than one child, and have overall monthly spending 5 × times more than Segment "0."

Segment "2" is the third largest segment, representing 22% of the market with the highest income (78 K). They spend almost 13 × times (622.8) on wines, with a frequency of 2.5 × times (20.6) compared to Segment "0." They were relatively older consumers (53 years), had a partner but no children, lived alone, and had the highest overall spending (768.72).

Segment "3" is the smallest segment, representing 21% of the market with the second lowest income (42 K). They spend almost 1.8 × times (87.8) on wines, with a frequency of 1.2 × times (10.1) compared to Segment "0." They were relatively older consumers (57

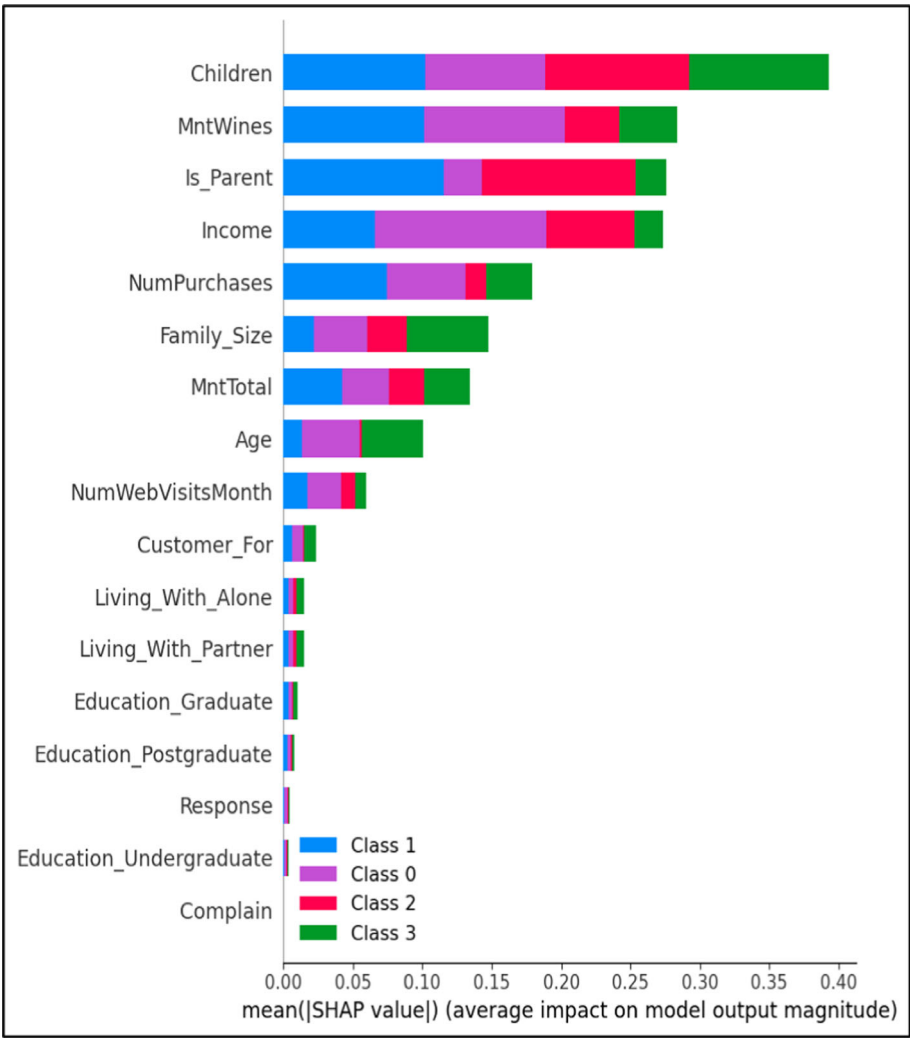


Fig. 5 Feature rank using Shapley Values – SHAP KernelExplainer Approach

years), had a partner and two children, and had the lowest overall monthly spending (65.74). As seen from the segment descriptions, the segments comprised relatively distinct customers in terms of the top 10 features (refer to Table 10) identified using XAI.

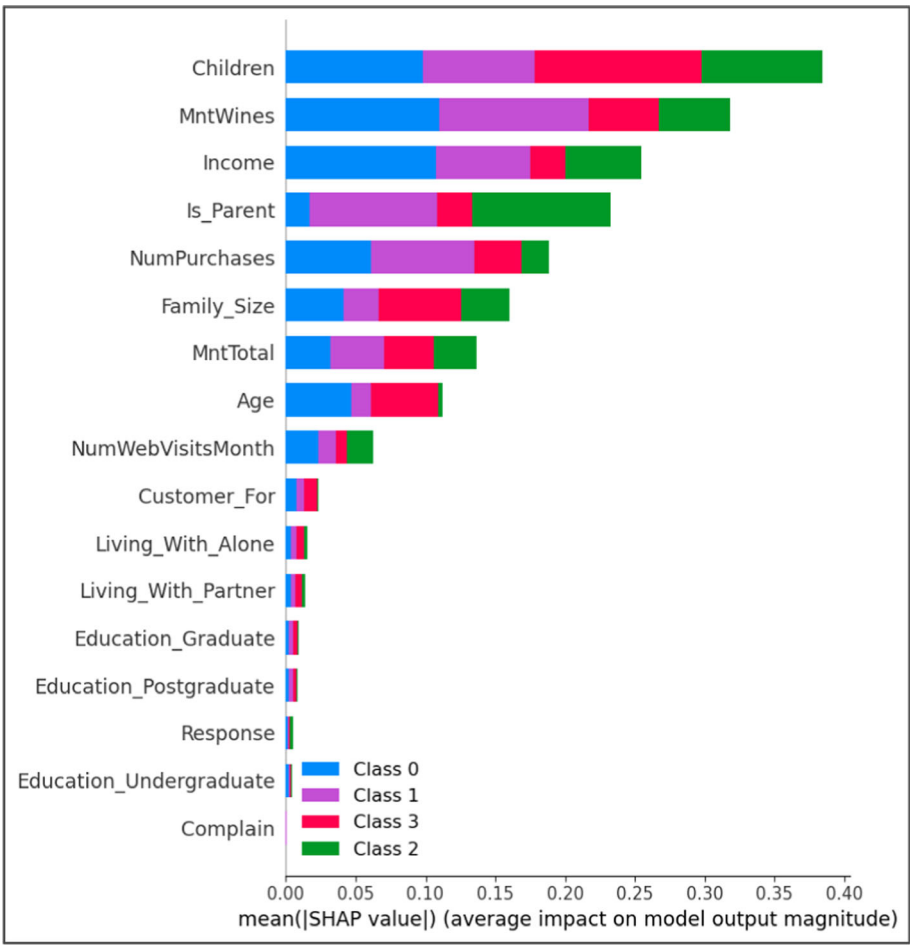


Fig. 6 Feature rank using Shapley Values – Tree Explainer Approac

5.4 Forecasting models and benchmarking

5.4.1 Baseline model

The baseline model is a naïve model that does not require significant memory or time and produces accurate results; therefore, we need a baseline model to compare our results. In the baseline model, the sales on the following day were the same. For instance, sales on days 1, 2, and 3 were 100, 200, and 300, respectively. In this case, the naïve model predicts day 2 sales as 100 and day 3 sales as 200. The baseline model had an RMSE of 25.859, 137.12, 246.664, and 73.314 for clusters 0, 1, 2, and 3, respectively, and the overall RMSE was 294.778 for the test dataset.

Table 10 Cluster Summary by Top '10' Distinguishing Features

Feature	Cluster "0"	Cluster "1"	Cluster "2"	Cluster "3"
Size	30.52%	26.81%	21.70%	20.97%
Children	0.77	1.16	0.01	1.91
MntWines	46.87	512.51	622.81	87.92
Income	31,329.84	62,244.59	77,508.53	42,392.49
Is_Parent	0.77	1.00	0.01	1.00
NumPurchases	8.43	21.64	20.25	10.13
Family_Size	2.41	2.81	1.61	3.59
MntTotal	84.03	357.11	768.72	65.74
Age	45.27	55.14	52.85	57.30
NumWebVisitsMonth	6.80	5.23	2.58	6.13
Customer_For	517.81	549.08	506.95	462.07

5.4.2 Autoregression (AR) model

AR models assume autocorrelation between the data in the initial and current timestamps. In the AR models, we used the previous values of the independent variable to predict its future values and referred to the variables involved as lag variables. The number of terms in the regression model increased with the number of lags. For simplicity, this model was constructed using only MntWines, which is the amount spent on wine. We determined the lag values to be two from the autocorrelation plot and then obtained the model. The autoregression model obtained is as follows: Li,i = 1,2 lag of 1 and 2, respectively, Mntwines = 35.094843 + 0.222446* MntWines.L1 + 0.272481* MntWines.L2. The AR model has an RMSE of 43.464, 179.008, 181.506, and 92.666 for Clusters 0, 1, 2, and 3, respectively, and the overall RMSE was 259.776 in the test dataset.

5.4.3 Vector autoregression (VAR)

While AR deals with a single time series, it can handle two or more time series that influence each other. Christopher Sims introduced VAR and used it to model causal relationships among macroeconomic variables. In VAR, the target variable depends not only on its previous values but also on the previous values of other variables. We first determined the suitability of the data for VAR and lag order using the Granger causality test. After identifying the variables that influence MntWines, we performed cointegration testing to determine whether the variables had a significant relationship. We applied the VAR model to train data after a test for stationarity (by Augmented Dickey-Fuller test) and identity lag order by selecting the model with the lowest AIC. The RMSE values for clusters 0, 1, 2, and 3 and the overall data are 40.681, 173.397, 184.361, 92.573, and 232.069, respectively.

5.4.4 Support vector regression (SVR)

SVR is a regression technique that uses the same principle as SVMs. SVRs are among the most potent ML techniques for regression, and their advantage is that they are not bound

by assumptions applied to parametric models. SVRs provide various kernels to model the nonlinear relationships among the variables. The SVR models yielded an RMSE of 28.61, 116.69, 141.46, 71.96, and 247.29 for clusters 0, 1, 2, and 3, respectively, and the overall data in the test data.

5.4.5 Univariate LSTM

LSTMs are a Recurrent Neural Network type of RNN used in the context of time-series data. LSTMs have specific memory cells in their architecture that store long-term dependencies (Hochreiter & Schmidhuber, 1997). LSTMs also overcome the vanishing gradient problems typically observed in DNNs used for sequential data. LSTMs also exhibit architectural variations: LSTM, Stacked LSTM (multiple LSTMs are stacked here), and BiDirectional LSTM. Furthermore, the LTSM architecture involves several hyperparameters, and we use the grid-search method to tune the hyperparameters to obtain the best result. Table 11 presents the test data results for various types of LSTMs across clusters.

5.4.6 Forecasting model selection

Table 12 summarizes the performance of the various forecasting models. As seen in Table 12, the Single-Layer LSTM showed superior forecasting performance in three clusters: Cluster 0, Cluster 2, and Cluster 3, and SVR showed better performance in Cluster 1. The performance improvements achieved by the Single-Layer LSTM over SVR were 13.28%, 85.86%, and 14.7% for Clusters 0, 1, and 3, respectively. However, Cluster 3 showed inferior performance (-12.85%) compared to SVR. Traditional approaches, such as AR and VAR, show inferior performance compared with SVR and LSTM. Therefore, we may conclude that ML performance offers superior predictive performance compared to traditional techniques for demand estimation.

6 Discussions and implications

This section discusses the major findings of this study and their managerial implications.

6.1 Discussions on results

The impact of perishable products on the environment is significant and multifaceted. Rapid spoilage of these products leads to waste throughout the supply chain, which ultimately ends up in landfills and generates methane, a potent greenhouse gas. Second, the packaging used to protect perishables during transportation and storage can create environmental issues if they are not properly managed or made from unsustainable materials. Finally, the energy required to maintain proper temperature control during transport and storage often relies on fossil fuels, which contributes to environmental degradation. To mitigate these environmental effects, efficient cold chain management is necessary to minimize spoilage and reduce the impact of perishable products(Shafiee et al., 2022). This study contributes to the literature in this regard.

The research objectives were to (1) identify customer segments through hierarchical clustering and apply XAI to describe the segments, (2) estimate wine consumption using various

Table 11 Train and test RMSE values for LSTM

Model	Cluster 0		Cluster 1		Cluster 2		Cluster 3		Overall	
Train/Test RMSE	Train	Test	Train	Test	Train	Test	Train	Test	Train	Test
Single LSTM layer	67.26	24.81	247.63	131.69	66.65	26.84	90.25	61.38	409.23	238.18
Stacked LSTM with 2 layers	67.41	21.30	247.76	140.76	283.07	176.32	92.15	64.67	407.79	223.61
Bidirectional LSTM	67.35	23.26	250.39	143.48	90.43	62.83	90.16	61.20	407.96	233.26
Grid search	68.04	32.68	257.17	177.48	288.77	182.58	92.05	68.47	424.67	232.26

Table 12 Test RMSE values for different models

Model	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Overall
Baseline	25.859	137.12	246.664	73.314	294.778
Autoregression	43.464	179.008	181.506	92.666	259.776
Vector Autoregression	40.681	173.397	184.361	92.573	232.069
SVR	28.61	116.69	189.89	71.96	247.29
Single-Layer LSTM	24.81	131.69	26.84	61.38	238.18
Best Model	LSTM	SVR	LSTM	LSTM	VAR

ML techniques for each of the identified clusters, and (3) compare the efficacy of ML techniques with conventional forecasting methods.

The first research objective is accomplished in Sect. 5.2, where we applied and compared k-means and hierarchical clustering techniques to generate clusters. In Sect. 5.3, we used surrogate models and XAI techniques to draw insights into the features that significantly differed among the clusters. The second research objective was achieved in Sect. 5.4, where we employed several ML techniques to predict the demand in each identified segment and the overall data. A key requirement of an XAI technique is its reliability. We followed an approach to establish reliability by comparing results across three XAI techniques: relative importance approach, model agnostic – SHAP, and model specific – SHAP Standard Python libraries–SHAPKernelExplainer and SHAPTreeExplainer, which were used for this purpose. We found that the top 5 contributing factors by SHAP approaches were 'Children, ' 'MntWines, ' 'income,' 'IS_Parent,' and 'NumPurchases.' From the results generated by XAI, we found 100% commonality among the Top-10 features between the three techniques and 80% among the Top-5 features determined by relative importance and SHAP techniques. Thus, we conclude that the proposed approach is reliable.

The third research objective was to compare the efficacy of machine-learning techniques with conventional forecasting methods. The results summarized in Table 12 show that ML models are superior to the conventional methods. We found that the Single-Layer LSTM model exhibited superior performance in three of the four segments. We now show whether the forecasting accuracy improves when predictions are made at the segment level rather than the entire dataset. This can be inferred by comparing the RMSE values for clusters 0, 1, 2, and 3 with the RMSE values for the entire dataset. As shown in Table 12, the RMSE values using the Single-Layer LSTM at the segment level were 24.81, 131.69, 26.84, and 61.38. However, the RMSE increases to 238.18 when forecasting is made for the entire dataset. The weighted average error (by segment size) across segments was 61.57, yielding an improvement of 74.14% over the forecasting using the entire dataset. Thus, the forecasting is more accurate at the segment level. The results validate the core premise of this study, which is that ML-based customer-centric segmentation and forecasting allows us to predict demand with higher accuracy.

6.2 Theoretical and managerial implications

This section provides theoretical and managerial implications in the following three subsections.

6.2.1 Theoretical implications

(a) XAI in supply chain resilience: This study enhances the XAI in supply chain management by focusing on the resilience of perishable goods. Comprehending AI decision-making is crucial for accountability and transparency. This study combines XAI with supply chain resilience, demonstrating how transparent AI models can forecast inventory requirements using customer characteristics and behavior patterns.

(b) Customer-centric inventory optimization: This study enhances supply chain intelligence using AI to incorporate customer characteristics into perishable goods inventory prediction, enabling stakeholders to understand the impact of these factors on inventory decisions.

(c) XAI in inventory optimization: This study demonstrates how XAI can bolster supply chain resilience for perishables, addressing the need for transparency in AI decision-making to ensure responsible AI practices.

(d) Customer-centric predictive models: Our research contributes to the predictive inventory management literature by integrating customer characteristics into AI models, thereby enhancing the accuracy and relevance of predictions for perishable products such as wine.

(e) Transparency and responsibility in AI: This study highlights the importance of responsible and transparent AI practices in operations research to ensure that AI-driven supply chain decisions are explainable, understandable, and trustworthy.

6.2.2 Managerial implications

This study extends the literature on the application of AI to inventory-management problems. Our study reinforces the application of ML to inventory management problems and is in line with Huber and Stuckenschmidt (2020). We extend the extant state-of-the-art by proposing a two-stage framework for demand forecasting.

This study has several practical implications. The first implication for managers is that inventory management problems may be decomposed into multiple ML tasks, each of which may be accomplished using the most suitable technique. The implication is that considering customer characteristics in the demand estimation approach improves forecasting accuracy. As shown in Sect. 6, forecasting errors are reduced by 74.14% at the segment level.

Second, we operationalized and validated a procedure for incorporating customer characteristics into the demand estimation process. As a result, managers can employ our approach, which sequentially integrates a descriptive ML task, namely clustering, and a predictive task, namely forecasting. Although we have not considered sophisticated techniques for clustering and forecasting, such as autoencoders and reinforcement learning, future studies may consider them.

Meske et al. (2022) established the objectives of XAI as to explain, evaluate, learn, and justify AI. The third implication is that we demonstrate how to achieve these four objectives of XAI in supply chain forecasting. We leveraged XAI to improve explainability and evaluate and learn from the results. We used the classification model to evaluate and conclude that the segments from the clustering were robust (classification metrics > 90%). Feature importance and Shapley values are explained and learned by extracting the features that contribute to distinguishing the segments, thus justifying clustering. Managers must adopt XAI to foster trust and AI interventions.

Nevertheless, it is crucial to thoroughly scrutinize the ethical and operational challenges inherent in AI. AI implementations should transcend the mere achievement of objectives such as explanation, evaluation, learning, and justification in solution development (Mesket

et al., 2022). Siau and Wang (2020) introduced a framework for constructing ethical AI solutions. Ethical AI requires a strategic approach that comprehensively prioritizes technological integrity and addresses various dimensions. These include (a) fostering transparency in both data and model; (b) ensuring stakeholder data security and privacy through measures such as data anonymization and adherence to data protection regulations; (c) mitigating human bias in data to ensure fairness and impartiality; (d) delivering responsible and accountable solutions that align with human rights laws, democratic principles, civil rights, and strike a balance between automation and job preservation; (e) continuous enhancement of models; (f) obtaining approval from ethics review boards; and (g) institutionalizing ethical frameworks advocated by organizations such as the United Nations (UN), Institute of Electrical and Electronics Engineers (IEEE), and the European Commission, among others.

Digital transformation affects the supply chain by enhancing its operational, structural, and strategic flexibility (Kopanaki et al., 2022). The digital transformation process involves leveraging information technology to sense and respond to challenges in the business environment. Managers should prioritize investment in digital transformation and establish an ecosystem of data warehouses, exploratory platforms, analytical tools, and visualization environments to support timely and precise decision making. Whereas digital transformation typically encompasses a variety of information technologies, organizations can consider adopting cloud computing-based services to implement AI solutions, making such solutions cost-effective and scalable.

Finally, as illustrated in the study, adopting Industry 4.0, such as ML, for data-driven decision-making would lead to efficiency gains. Improved demand forecasting is the first step toward improving inventory management practices. Managers may use these results to reduce inventory carrying costs due to overstocking, prevent opportunity losses due to stockouts, adjust production planning operations in wineries, reduce variance in transmitting demand upstream, and reduce waste at the farm level. We compared ML and traditional forecasting techniques and established their superiority in demand estimation for perishable products.

Forecasting the demand is significant for perishable goods that cannot be consumed after a certain period. We analyzed wine data because grapes are perishable products, for which accurate forecasting is essential. As the quality of the grapes determines the quality of wine, the quality of grape wine affects how the company perceives it and the price at which wine can be sold. As the wine industry in India is still in its infancy, accurate demand forecasting is essential for reducing inventory costs. Our approach can assist in reducing waste, which in turn reduces costs; additionally, this approach can be used in other industries involving perishable products, such as the fruit juice, pickle, and dairy products industries, to reduce the cost of holding inventory (which involves storing them in cold storage that is expensive) and reduce the cost of waste (perishable products become hazards quickly and cannot be used in production processes).

The wine industry is complex and requires a deep understanding of consumer preferences, behavior, and characteristics. This study delves into the nuances of demand forecasting in this sector, examines its challenges, attempts to contribute by exploring the factors affecting consumer behavior in the wine category, and highlights the obstacles forecasters face when predicting consumer demand for wine.

The wine industry encompasses the entire grape cultivation process, wine production, distribution, and marketing. Consumer preferences play a crucial role in this industry, with factors such as (a) demographics, age, income, and cultural background; (b) tastes and preferences, including preferences for grape variety, sweetness level, and region; and (c) evolving trends, including sustainability and ethical sourcing (Fazio et al., 2023; Volpe et al., 2010).

Wine demand forecasting is complex for several reasons, including (a) seasonality, with sales often surging during holidays and on specific occasions; (b) vintage variation and weather conditions significantly affecting grape quality and overall vintage appeal (Volpe et al., 2010); (c) economic factors, with economic downturns leading to decreased spending on non-essential items such as wine; (d) new product launches, with the introduction of new varieties or brands disrupting established preferences; and (e) consumer behavior, with consumer tastes and preferences being complicated and influenced by various factors such as marketing and critical reviews. These factors make it challenging to predict future demand accurately (Fazio et al., 2023).

Consumer behavior in the wine sector is complex and multifaceted, contributing to the difficulty in accurate demand forecasting. This complexity arises from (a) subjective preferences, with wine appreciation being subjective and influenced by emotional and social aspects beyond price and quality; (b) evolving preferences, with consumer preference shifting owing to exposure to new wines, trends, or recommendations; and (c) external influences, with unexpected events, such as weather disasters or economic crises, significantly impacting consumer buying behavior. Therefore, wine demand forecasting requires a nuanced approach that considers historical sales data and qualitative factors affecting consumer preferences (Mehta & Bhanja, 2017; Stanco et al., 2020).

Digital transformation can address the challenges mentioned in the previous section and improve the resilience of perishable product supply chains. This study aims to improve the forecasting accuracy using customer segmentation and ML techniques for sales forecasting, such as hierarchical clustering and support vector regression. Consequently, the proposed technique emphasizes customer-centric demand forecasting. It lowers forecasting error more than traditional forecasting techniques, which rely solely on sales data and place limited emphasis on customers. This research focuses on the wine industry, which has received little attention, particularly in India, and is still in its early stages. However, this study can also be applied to other industries that produce perishable goods. Furthermore, we demonstrated a method to incorporate XAI as part of the machine learning process, which can easily be adopted in any supply chain problem requiring AI intervention. We adopted three XAI techniques: (a) conducting exploratory data analysis and employing dimensionality reduction to improve data explainability; (b) using feature importance for feature-based explainability; and (c) employing kernelShap for example-based explainability to better understand data, features, and segment characteristics. The outcomes confirm the effectiveness of employing segment-specific forecasting and encouraging XAI adoption. Future investigations could explore other XAI techniques, such as counterfactuals, anchors, and contrastive explanation methods (Dwivedi et al., 2023).

7 Conclusions limitations and directions for future research

This section presents the conclusions, limitations, and directions for future research.

7.1 Conclusions

This paper presents a framework for optimizing inventory levels and reducing forecasting errors in perishable supply chains in two steps: customer segmentation using k-means, hierarchical clustering, and XAI techniques, followed by demand forecasting for each segment using transparent AI methods. LSTM outperforms autoregressive models and support vector

regression for demand forecasting. The two-step segmentation and demand forecasting processes surpassed the traditional methods that did not incorporate segmentation. This study underscores the use of XAI to enhance transparency in machine learning and improve supply chain resilience for perishable products. Specifically, the framework involves customer segmentation and demand forecasting using XAI techniques, employing k-means clustering, hierarchical clustering, and XAI to compare machine learning methods for demand forecasting. The results indicate that hierarchical clustering and LSTM in a two-step process outperformed conventional unsegmented forecasting. This approach demonstrates XAI's role of XAI in improving transparency and reducing forecasting errors, thereby enhancing the resilience of perishable product supply chains, notably in the Indian wine industry.

The primary novelty of this study is the application of XAI to increase machine learning transparency and minimize forecasting errors in real-world scenarios, which is especially important for Operations Research (OR) applications that require explainability and trust in decision-making. Enhanced demand forecasting can improve the resilience of supply chains to perishable goods.

- Optimizing costs and profits while meeting customer demand is a key challenge in perishable supply chains.
- This study proposes a two-step solution: customer segmentation and demand forecasting using XAI.
- K-means, hierarchical clustering, and XAI were employed to segment, model, and compare the machine learning techniques.
- LSTM outperforms autoregressive models and support vector regression for demand forecasting.
- The two-step segmentation and demand forecasting process outperforms the conventional single-step unsegmented forecasting.

7.2 Limitations and directions for future research

The proposed method presupposes that the sales data contain inherent customer segments. This study is limited because customer data are required to perform a major task: identifying customer segments. However, most businesses lack access to this data. Second, the data we used were from 2015, and the segment characteristics could have changed over the years. The algorithms used for customer segmentation and demand forecasting may vary depending on the dataset. Obtaining this information is particularly challenging in developing countries like India, where most perishable product purchases are made in small brick-and-mortar stores. Furthermore, forecasting techniques require data collection at regular intervals. However, our research began with real-world data to create customer segments and realistic data to meet forecasting requirements. The results underscore the importance of embracing and examining digital technologies across various domains for enhanced information collection (e.g., IoT), dissemination (e.g., pervasive networks), processing (e.g., cloud computing and big data), management (e.g., AI), and security (e.g., blockchain).

Based on our findings, further research on supply chain resilience is necessary. Small- and medium-sized enterprises' perishable product supply chains are critical to the global economy's survival, but they face enormous challenges due to the nature of their operations. Given the importance of perishable product supply chains in sustaining fresh food supply and the global economy, future research could simultaneously focus on developing and implementing resilient supply chain designs and recovery strategies. The impact of disruptions on

supply chains, resilience methods in supply chain design, and recovery strategies supported by cost–benefit analysis would all be commendable additions.

Acknowledgements The authors have not received any funds for this research work and totally depend upon themselves to carry out this study.

Funding Open Access funding enabled and organized by CAUL and its Member Institutions.

Data availability All data have been included in this paper.

Declarations During the preparation of this work, the author(s) used [Grammarly and Paperal] to improve the manuscript's grammar. After using this tool/service, the author(s) reviewed and edited the content as needed and took (s) full responsibility for the publication's content**.**

conflicts of interest The authors have no conflicts of interest to declare.

Ethical approval This article contains no studies involving human participants or animals performed by authors.

Appendix A See Table 13.

Table 13 Models and configuration

No	Model	Configuration
1	Hierarchical Clustering	AgglomerativeClustering n_clusters = 4 affinity = "euclidean" linkage = "ward"
2	Random Forest	max_depth = 11, n_estimators = 500
3	LSTM	Number of Layers = 1 Number of LSTM units = 4 Optimizer = 'adam' Batch Size = 1 Epochs = 30
4	SVR	Kerbel = 'rbf'
5	Autoregression	lags = 2; seasonal = False
6	Vector Autoregression	maxlags = 7
7	Feature Importance Using SHAP	TreeExplainer

Appendix B: Description of techniques used Hierarchical clustering

Clustering methods belong to the category of unsupervised machine learning. Clustering aims to identify patterns and group the data using its features. Based on their approach, clustering techniques can be classified into partitioning, hierarchical, density, and grid-based methods. This study used k-means (partition method) and hierarchical clustering (hierarchical method). Partitioning algorithms divide the given "n" data points into "k" partitions or, in other words, "k" clusters. The objects are clustered such that the objects in a cluster are similar and dissimilar to objects in other clusters. Hierarchical clustering algorithms create top-down or bottom-up decompositions of data points. In the top-down approach, a data cluster is progressively decomposed into smaller clusters, whereas, in the bottom-up approach, individual data points are progressively agglomerated into a larger cluster. As shown in Appendix A, agglomerative clustering was used in the configuration.

Random forest

Random forests belong to the class of ensemble models used in machine learning. Ensemble models use a set of weak learners to build a robust model. A random forest is an ensemble of multiple decision trees, where

each decision tree is trained on a randomly chosen subset of data features. The outcomes of the decision trees were aggregated to arrive at the outcome. The Random Forest model used in this study had a maximum depth of 11 and 500 estimators.

LSTM

LSTMs is a recurrent neural network (RNN) class that uses memory cells to handle long-term dependencies. LSTM consists of input, output, and forget gates to selectively retain, modify, and discard long-term information, respectively. The LSTM processes the inputs sequentially at each time step, and the operations are as follows:

Forget Gate: Determines the parts of the memory that have to be forgotten using the previous hidden state and current inputs. The forget gate vector f_t is given by $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$, where σ, W_f, x_t, h_{t-1}, and b_f are the sigmoid activation function, weight matrix of the forget gate, current input, previous hidden state, and bias term of the forget gate.

Input Gate: Determine the information to update the current cell state. It determines the input gate vector i_t and the candidate cell state. $\widetilde{C}_t$. The input vector i_t is given by $i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$ and $\widetilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$ where tanh, W_i, W_c, b_i, and b_i are hyperbolic tangent activation function, weight matrixes, and bias terms, respectively.

Cell State Update: Information from the forget and input gates is used to update the cell state. The new cell state C_t is given by $C_t = f_t \odot C_{t-1} + i_t \odot \widetilde{C}_t$.

Output Gate: Determine the hidden state h_t for the next time step. The hidden state h_t is given by $h_t = O_t \odot \tanh(C_t)$ and $O_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$, where W_o and b_o are the weight matrix and the bias term for the output gate, respectively.

As shown in Appendix A, a single-layer LSTM configuration was used.

SVR

Support Vector Machines belong to the family of boundary separation methods used in machine learning. SVMs can be used for regression (SVR) and classification (SVC). SVR aims to identify the optimal hyperplane within threshold ϵ for any given y. SVR can be applied to linear and nonlinear regressions. SVR supports nonlinear regression through popular kernel functions, such as RBF, Polynomial, and Sigmoid. In this study, we applied the RNF kernel while implementing the SVR.

Autoregression

Autoregression (AR) means the regression of a variable measured at different periods. This difference in period is termed the lag. The equation of AR with lag p, i.e., AR(p), can be represented as $Y_{t+1} = \beta_1 Y_t + \beta_2 Y_{t-1} + \ldots + \beta_p Y_{t-p+1} + \varepsilon_{t+1}$. The number of lags was identified from the partial autocorrelation plots. In this study, we used an Auto Regression Model with lag 2.

Vector autoregression

Unlike Autoregression, A VAR model uses multiple time series variables. Each variable in the system was regressed on past values of the variables in the system. We modeled the VAR with the following variables: MntWines, MntTotal, NumWebVisitsMonth, NumPurchases, Age, Children, Family_Size, Education_Graduate, Education_Postgraduate, Living_With_Alone, and Living_With_Partner.

Feature importance of using shap

Shapley valued quantity as the impact of a coalition of players in a cooperative game. In machine learning, the model is a coalition of features and the output is the payoff. The Shap values calculate the difference between the actual and average predicted outputs for different coalitions with and without a feature. The difference provides the marginal contribution of a feature across all coalitions (subsets of features). A positive/negative Shap value indicates that the feature positively/negatively affects the prediction. The magnitude of the Shap value indicates the strength of the impact of a feature.

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